

CS & IT ENGINEERING



Computer Network

Transport Layer

Lecture No. - 01



By - Abhishek Sir



Recap of Previous Lecture



Topic

Ethernet

Topic

Back-off Time

Topic

Ethernet Bridging



Topics to be Covered



Topic

Transport Layer

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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#Q. A and B are the only two stations on an Ethernet. Each has a steady queue of frames to send. Both A and B attempt to transmit a frame, collide, and A wins the first backoff race. At the end of this successful transmission by A, both A and B attempt to transmit and collide. The probability that A wins the second backoff race is:

[GATE 2004]

(A) 0.5

✓ (B) 0.625

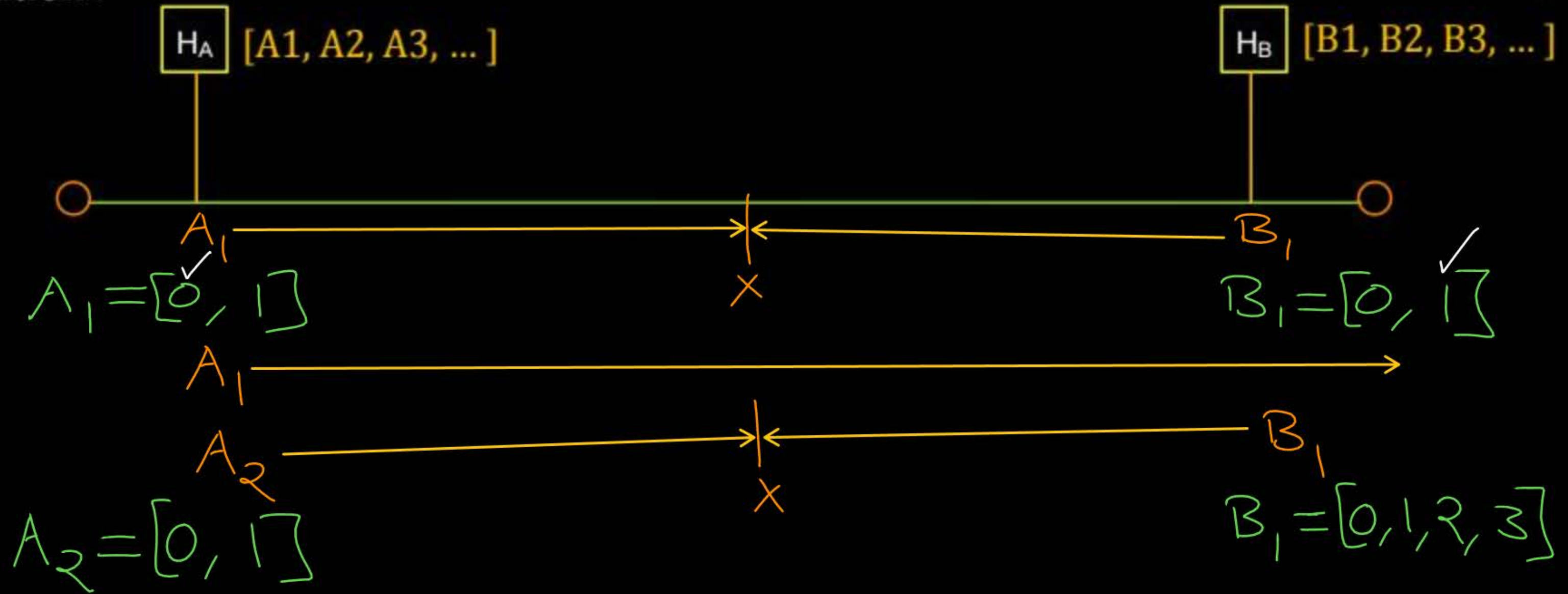
(C) 0.75

(D) 1.0

$$\text{Ans} = \frac{5}{8}$$

Ans: B

Solution :-



A2	B1	Result
0	0	} Again collided
1	1	
0	1	} A win
0	2	
0	3	
1	2	
1	3	} B win
1	0	

[CSMA/CD]
#Q. In an Ethernet local area network, which one of the following statements is TRUE?

[GATE 2016]

- ☒ (A) A station stops to sense the channel once it starts transmitting a frame. FALSE
- ☒ (B) The purpose of the jamming signal is to pad the frames that are smaller than the minimum frame size. FALSE
- ☒ (C) A station continues to transmit the packet even after the collision is detected. FALSE
- ☒ (D) The exponential backoff mechanism reduces the probability of collision on retransmissions. TRUE

Ans: D

#Q. Which of the following statements is TRUE about CSMA/CD?

[GATE 2005]

- ☒ (A) IEEE 802.11 wireless LAN runs ^{CSMA/CA} CSMA/CD protocol. FALSE
- ☒ (B) Ethernet is not based on CSMA/CD protocol. FALSE
- ☒ (C) CSMA/CD is not suitable for a high propagation delay network like satellite network. TRUE
- ☒ (D) There is no contention in a CSMA/CD network. FALSE

$$\text{Contention slot size} = (2t_p + t_{xj})$$

Ans: C

#Q. Which of the following statements is FALSE regarding a bridge?

[GATE 2005]

(A) Bridge is a layer 2 device TRUE

(B) Bridge reduces collision domain TRUE

(C) Bridge is used to connect two or more LAN segments TRUE

✓ (D) Bridge reduces broadcast domain FALSE

Ans: D

#Q. In a network of LANs connected by bridges, packets are sent from one LAN to another through intermediate bridges. Since more than one path may exist between two LANs, packets may have to be routed through multiple bridges. Why is the spanning tree algorithm used for bridge-routing?

[GATE 2005]

- ☒ (A) For shortest path routing between LANs
- ☒ (B) For avoiding loops in the routing paths
- (C) For fault tolerance
- (D) For minimizing collisions

Ans: B



Topic : Syllabus



Concept of layering : OSI and TCP/IP Protocol Stacks;

Basics of packet, circuit and virtual circuit switching;

Data link layer : framing, error detection, Medium Access Control, Ethernet bridging;

Routing protocols : shortest path, flooding, distance vector and link state routing;

Fragmentation and IP addressing, IPv4, CIDR notation, Basics of IP support protocols (ARP, DHCP, ICMP), Network Address Translation (NAT);

Transport layer : flow control and congestion control, UDP, TCP, sockets;



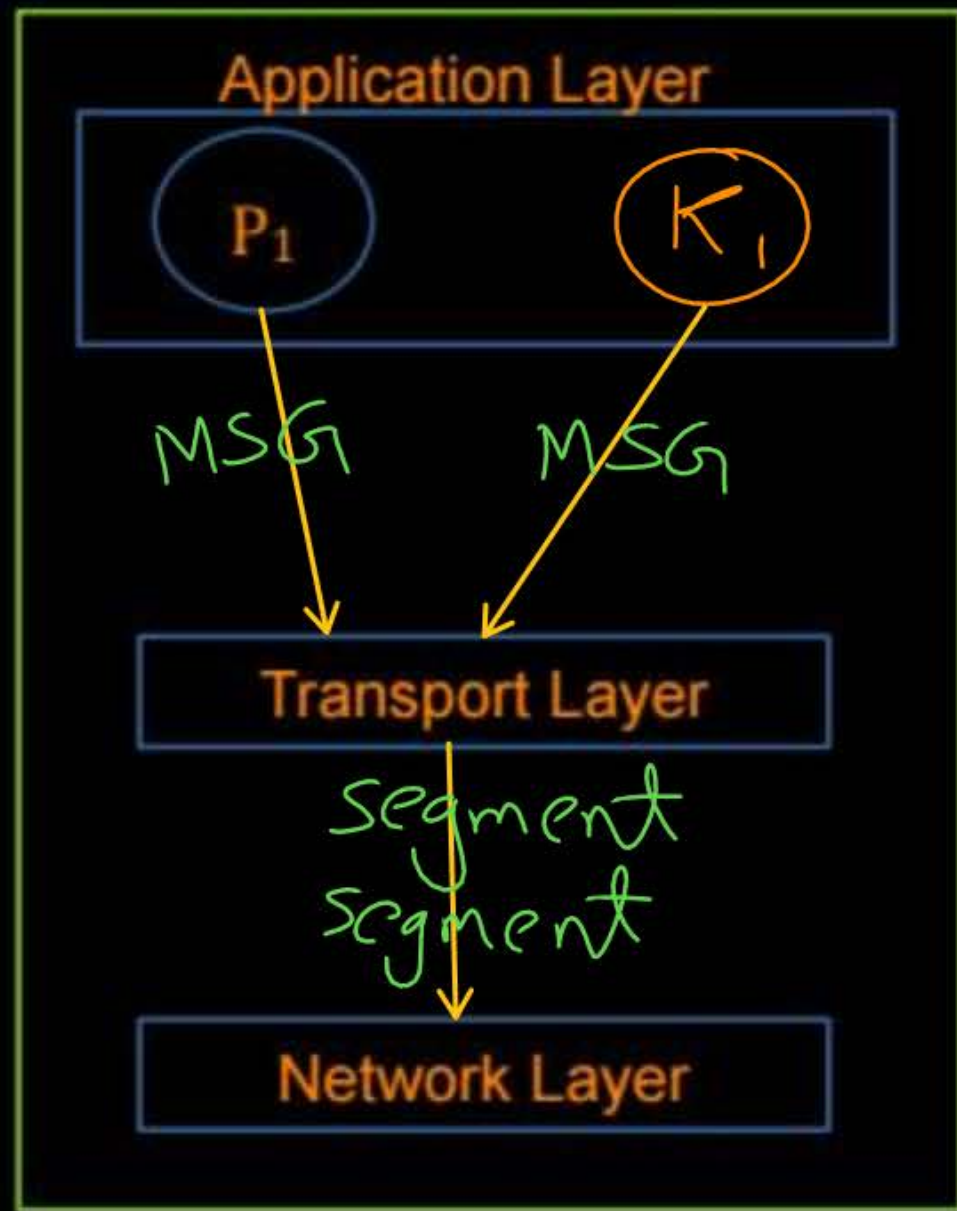
Topic : Transport Layer



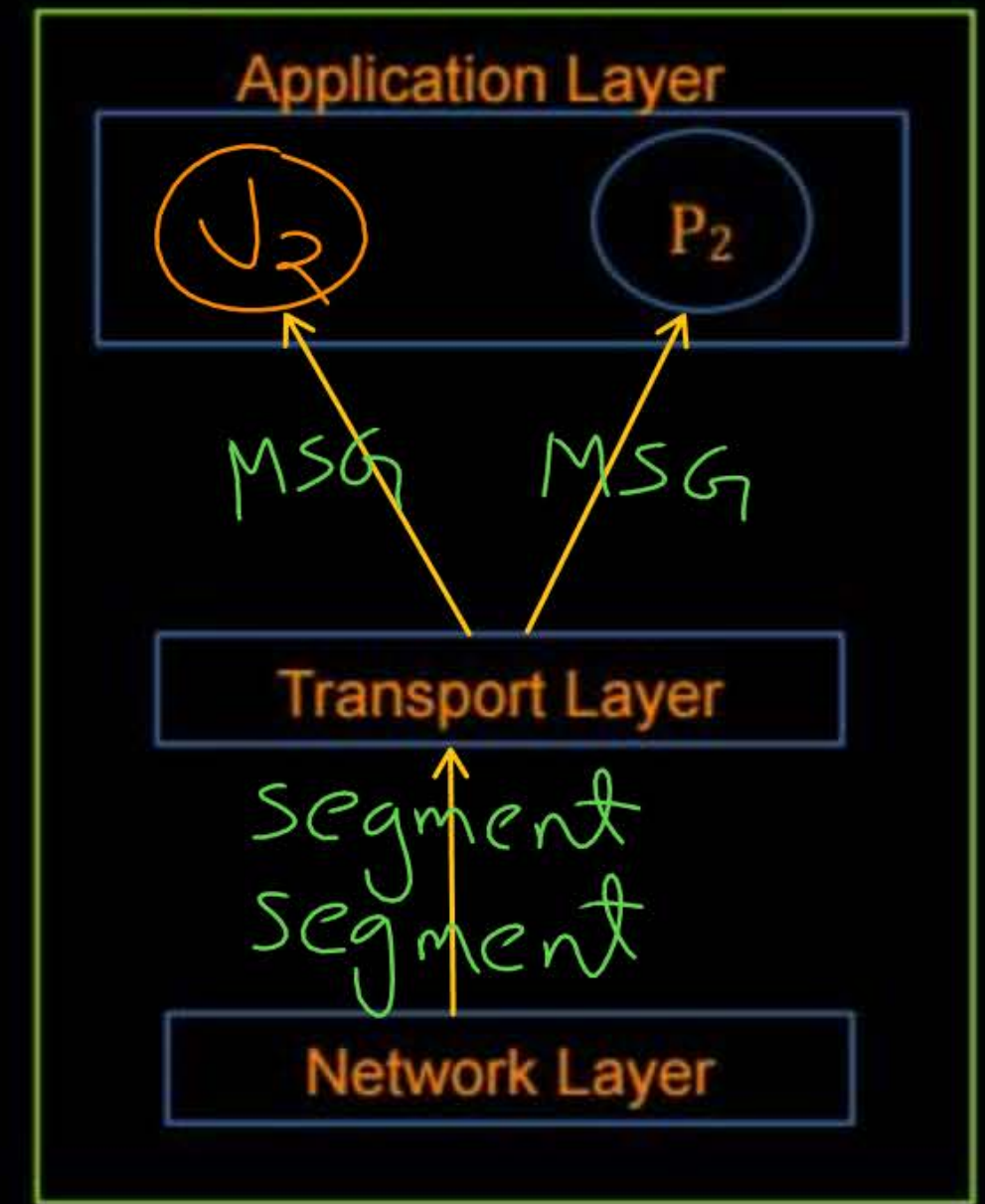
- Provide logical communication between application processes
(Processes running on different hosts)
- Responsible for process-to-process (end-to-end) communication
- Multiplexing & Demultiplexing

K_3 | J_4

Source Host



Destination Host





Topic : Transport Layer



→ Transport Layer PDU : "Segment"

→ Sender : Divide application messages into segments,
Segments passes to network layer

→ Receiver : Resembles segments into messages,
Messages passes to application layer



Topic : Socket



- One endpoint of a two-way communication
- Two network process communicate via a pair of sockets
- Door or interface of a network process
- Data structure



Topic : Multiplexing



Multiplexing at sender :

- > Handle data from multiple sockets
- > Add transport layer protocol header
[Used for demultiplexing at receiver)]



Topic : Demultiplexing



Demultiplexing at receiver :

1. Connection-less demultiplexing

→ UDP sockets identified by 2 tuples :

Destination IP address and Destination Port Number

2. Connection-oriented demultiplexing

→ TCP sockets identified by 4 tuples :

Source and Destination IP address

Source and Destination Port Number



Topic : Transport Layer



Two Transport Layer Protocols :

1. UDP : User Datagram Protocol $\Rightarrow$ Basic Protocol
2. TCP : Transmission Control Protocol

$$TCP = [UDP + Extra Services]$$



Topic : Socket





Topic : Socket





Topic : Transport Layer



Application Layer Protocol	Transport Layer Protocol
[DNS]	<u>UDP</u> (<u>default</u>), <u>TCP</u> (<u>conditional</u>)
<u>HTTP 1</u> & <u>HTTP 2</u>	[TCP]
<u>HTTP 3</u>	UDP
<u>FTP</u> , <u>SMTP</u> , <u>IMAP</u> (<u>POP</u>)	TCP
<u>DHCP</u>	UDP
RIP	UDP
BGP	TCP



Topic : Socket



=> Socket is uniquely identified by a combination of three factors :

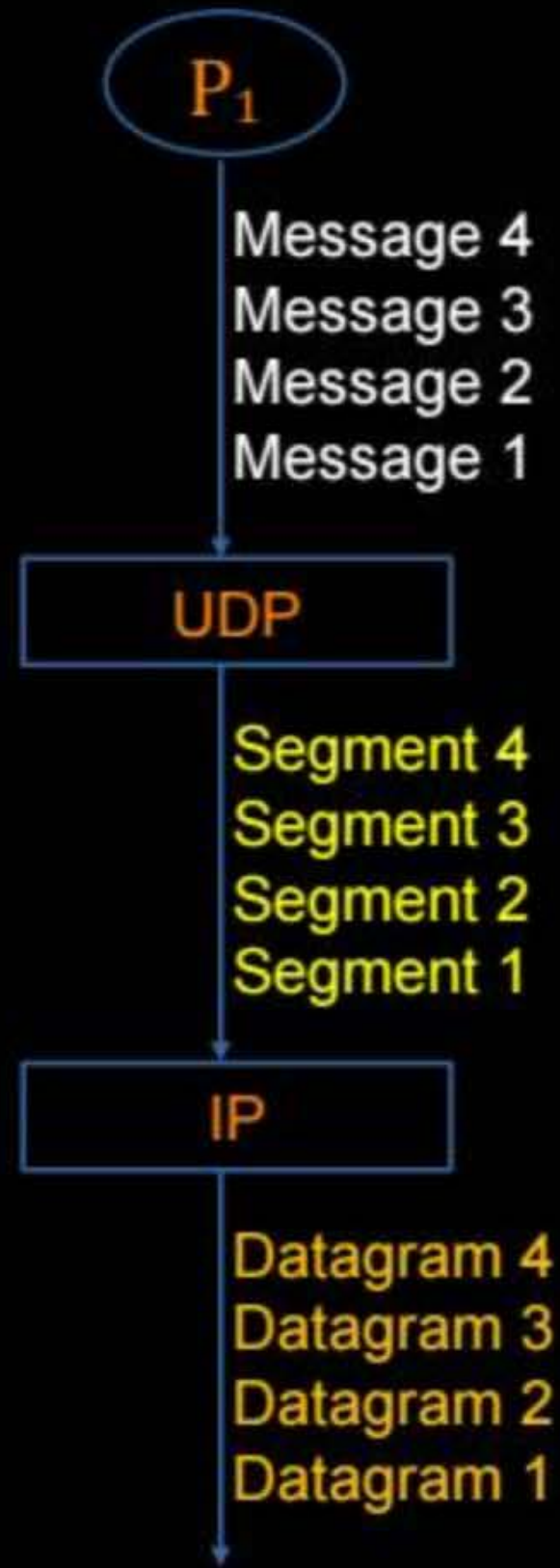
- > IP address
- > Port number
- > Transport protocol *



Topic : UDP



- UDP : User Datagram Protocol
- Provide 'Connection-less' and 'Unreliable' services $\Rightarrow IP$
[Unordered delivery of messages, messages may be lost]
- Prefer for shorter communication [like query and response]
- UDP as transport protocol used by :
DNS, SNMP, HTTP/3, RIP and Real Time Multimedia [Streaming Multimedia]





2 mins Summary



Topic

UDP

Topic

TCP



THANK - YOU

